

Ideological Bias in LLMs’ Economic Causal Reasoning

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Abstract

Do large language models (LLMs) exhibit systematic ideological bias when reasoning about economic causal effects? As LLMs are increasingly used in policy analysis and economic reporting, where directionally correct causal judgments are essential, this question has direct practical stakes. We present a systematic evaluation by extending the EconCausal benchmark with ideology-contested cases—instances where intervention-oriented (pro-government) and market-oriented (pro-market) perspectives predict divergent causal signs. From 10,490 causal triplets (treatment–outcome pairs with empirically verified effect directions) derived from top-tier economics and finance journals, we identify 1,056 ideology-contested instances and evaluate 20 state-of-the-art LLMs on their ability to predict empirically supported causal directions. We find that ideology-contested items are consistently harder than non-contested ones, and that across 18 of 20 models, accuracy is systematically higher when the empirically verified causal sign aligns with intervention-oriented expectations than with market-oriented ones. Moreover, when models err, their incorrect predictions disproportionately lean intervention-oriented—and this directional skew is not eliminated by one-shot in-context prompting. These results highlight that LLMs are not only less accurate on ideologically contested economic questions, but systematically less reliable in one ideological direction than the other, underscoring the need for direction-aware evaluation in high-stakes economic and policy settings.

1 Introduction

Do Large Language Models exhibit systematic ideological bias when reasoning about economic causal effects? This question is pressing because LLMs are increasingly deployed in high-stakes analytical workflows, such as economic reporting, policy evaluation, and corporate decision support (Kwon et al., 2024; Ludwig et al., 2025; Handler et al., 2024), where predicting causal directions correctly is essential. Yet in many such settings, that direction is genuinely contested: a single intervention can trigger competing mechanisms whose relative magnitudes are debated along ideological lines. A minimum wage increase, for instance, is expected to reduce hiring under a market-oriented framework that prioritizes labor cost effects, but to sustain employment under an intervention-oriented framework that emphasizes demand or job-matching gains. When LLMs predict causal directions in such settings, their outputs may implicitly privilege one set of priors over another, producing an invisible tilt in downstream policy advice.

Existing approaches to detecting such effects fall short in complementary ways. Work on LLM bias typically relies on survey-style instruments that probe expressed opinions or

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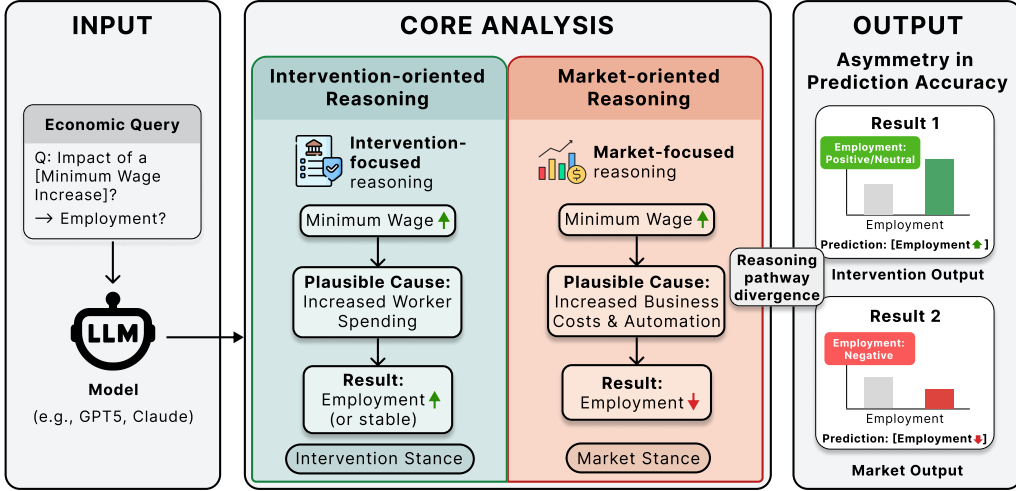


Figure 1: Overview of ideological divergence in LLM economic causal reasoning

preferences, often in social or political domains, and does not assess how models reason about causal relationships grounded in empirical evidence. In contrast, benchmarks for causal reasoning focus on formal logic or context-dependent inference, but assume a single correct expectation for each causal effect and do not account for settings where competing, internally consistent frameworks imply different expected directions. As a result, we lack a framework for asking whether LLMs reason about economic causation more reliably in one ideological direction than another, the setting where bias would have the most direct policy consequences.

In this paper, we present the first systematic analysis of ideological bias in LLMs’ economic causal reasoning. Leveraging the EconCausal benchmark dataset—a large-scale corpus of 10,490 context-annotated causal triplets derived from top-tier economics and finance journals—we identify 1,056 ideologically contested triplets (10.1%) where market-oriented and intervention-oriented frameworks predict divergent causal signs. We evaluate LLM accuracy separately across contested and non-contested settings, and define a directional bias measure that captures systematic asymmetries in model predictions across competing frameworks.

Through an extensive evaluation of 20 state-of-the-art LLMs, we document a striking asymmetry in reasoning ability. Models exhibit lower accuracy on ideologically contested triplets and perform significantly better when the empirical truth aligns with intervention-oriented expectations, showing an average accuracy gap of +9.7 to +15.1 percentage points relative to market-oriented cases. Errors are not random but systematically directional, and this pattern is robust across model families and difficulty levels. These findings highlight substantial reliability risks when LLMs serve as intermediaries in policy analysis, where implicit ideological anchoring can lead to systematically skewed conclusions.

Our contributions are as follows. First, we introduce the notion of ideologically contested causal relationships as a new lens for studying LLM causal reasoning beyond context dependence. Second, we construct a labeled subset of EconCausal distinguishing contested from non-contested triplets, enabling controlled evaluation across these settings. Third, we provide a systematic empirical analysis showing lower accuracy and asymmetric performance in contested cases. Fourth, we define a directional bias measure and document consistent asymmetry patterns across a range of models. Together, our results highlight the importance of evaluating causal reasoning not only across contexts, but also across settings where disagreement over causal expectations is intrinsic.

2 Related Work

2.1 Political and Ideological Bias in LLMs

Whether LLMs exhibit systematic political biases has attracted considerable attention. A common approach applies political orientation instruments to LLMs and positions their responses on ideological scales, and the emerging consensus is that instruction-tuned LLMs lean left on social and cultural dimensions (Santurkar et al., 2023; Rozado, 2024; Hartmann et al., 2023; Motoki et al., 2023). More recent efforts confirm a leftward tendency but show that its magnitude was overstated by earlier instruments, and that small changes in question format can substantially shift measured positions (Röttger et al., 2024; Faulborn et al., 2025).

These leanings have been traced to multiple sources. Feng et al. (2023) show that political biases in pretraining corpora propagate into downstream task predictions. Fulay et al. (2024) find that optimizing for truthfulness amplifies left-leaning tendencies in larger models, and Exler et al. (2025) document a monotonic relationship between parameter count and alignment with left-leaning parties.

However, these studies focus on expressed preferences in response to opinion-like questions. They reveal what positions LLMs appear to endorse, but not whether ideological priors systematically distort reasoning about verifiable factual claims. This distinction may be especially important in economic settings, where users can easily interpret a model’s causal prediction as reflecting an empirical regularity rather than a prior absorbed during training. Whether ideological leanings also shape how LLMs reason about economic causal relationships—where ground-truth effect signs are available from peer-reviewed empirical research—remains unexplored.

2.2 LLMs in Economic and Causal Reasoning

Work on causal reasoning suggests that LLMs often rely on priors rather than principled inference, which may help explain why directional bias can arise in economic prediction. Benchmarks such as CLadder and CORR2CAUSE consistently find that LLMs struggle with genuine causal reasoning across different levels of formalism (Jin et al., 2023; 2024; Wang, 2024; Chen et al., 2024; Yu et al., 2025). A common explanation is that their causal judgments rely heavily on priors absorbed during pretraining rather than principled inference: LLMs have been described as “causal parrots,” shown to exploit textual metadata rather than observational evidence, to adopt rigid rule-like strategies, and to inadequately revise beliefs when confronted with contradictory evidence (Zečević et al., 2023; Kıcıman et al., 2023; Wilie et al., 2024; Dettki et al., 2026).

Related work on LLMs in economic and policy settings points to similar limitations. LLMs have already been applied to policy-adjacent tasks such as classifying central bank communications (Silva et al., 2025), while existing benchmarks show persistent weaknesses in economic and causal reasoning (Raman et al., 2024; 2025; Quan & Liu, 2024; Guo & Yang, 2024). Yet this literature evaluates performance mainly in terms of accuracy and robustness, not whether prediction errors are systematically directional along ideological lines. Our work addresses this gap by testing whether LLMs exhibit ideologically asymmetric errors when predicting economic causal effect signs documented in the empirical literature.

3 EconCausal and Ideology-contested Extension

3.1 EconCausal Benchmark Overview

This study leverages the experimental framework of the EconCausal benchmark (Lee et al., 2026), which provides a dataset of causal relationships extracted from empirical studies in top-tier economics and finance journals. For each data instance, a causal triplet (T, O, Sign) consisting of the treatment T , outcome O , and the directional sign of the effect Sign was extracted, along with a corresponding context paragraph C that provides the essential socio-economic background for the observed relationship.

The primary experimental setup adopts a sign prediction task, requiring language models to predict the direction of the causal effect Sign given a context C and a treatment-outcome pair (T, O) as inputs. The sign label space is defined as

$$\mathcal{S} = \{+, -, \text{None}, \text{Mixed}\},$$

where the four classes represent significantly positive, significantly negative, insignificant, and heterogeneous/complex effects, respectively.

3.2 Defining Ideology-contested Causal Triplets

Following the EconCausal benchmark, each of the 10,490 causal triplets is assigned a ground-truth sign based on the original authors’ empirical specification. To construct ideology-conditioned labels, we queried four LLMs (GPT, Claude, Qwen, and Grok) once per triplet, asking each model to jointly assign an intervention-oriented expected sign and a market-oriented expected sign while remaining blinded to the paper’s empirical result. Detailed prompt is provided in Appendix E.

These signs are defined along the economic dimension of ideology (Alesina & Giuliano, 2011; Feldman & Johnston, 2014): $s^{\text{intervention}}$ reflects the expected causal sign under an intervention-oriented perspective that favors active government intervention to correct market failures, reduce inequality, and expand social insurance, while s^{market} reflects expectations under a market-oriented perspective that favors market-determined outcomes, individual responsibility, and limited government intervention.

We retained a triplet for ideology-contested labeling only when at least three of the four models assigned different signs to the two perspectives. For the retained triplets, we determined the final $s^{\text{intervention}}$ and s^{market} by majority vote across models for the intervention- and market-oriented signs, respectively. A triplet was then classified as ideology-contested when these final labels differed.

Formally, the ideology-contested subset is defined as

$$\mathcal{D}_{\text{div}} = \left\{ x \in \mathcal{D} \mid s^{\text{intervention}}(x) \neq s^{\text{market}}(x) \right\}. \quad (1)$$

This procedure yields 1,056 ideology-contested triplets (10.1% of the full set). Among these, 751 have a ground-truth sign coinciding with either $s^{\text{intervention}}$ or s^{market} , broken down:

- **Intervention-truth** (436 triplets, 58.1%): the ground-truth sign matches $s^{\text{intervention}}$.
- **Market-truth** (315 triplets, 41.9%): the ground-truth sign matches s^{market} .

Expert Validation. Two economics professors independently reviewed a representative subset of 126 triplets (18 per subcategory $\times$ 7 subcategories, balanced across intervention-oriented, market-oriented, and neutral alignments). Individual expert accuracies were 93.60% and 98.35%, respectively, and inter-rater agreement reached 93.33%, confirming the stability of the labeling process. The subfield composition of the ideology-contested subset, based on JEL classification codes, is detailed in Appendix A.

4 Main Results: Ideological Distortion in Economic Causal Reasoning

4.1 Evaluation Settings

We evaluate a diverse set of LLMs spanning both closed-source and open-source families. The closed-source group includes OpenAI GPT, Anthropic Claude, Google Gemini, and xAI Grok; the open-source group includes Meta Llama and Alibaba Qwen, chosen across varying parameter scales to examine the effect of model size.

For evaluation, we measure overall accuracy as well as accuracy on the ideology-contested and non-contested subsets separately. Within the ideology-contested subset, we define intervention-truth accuracy ($\text{Acc}^{\text{intervention}}$) as the accuracy on items whose ground truth aligns with intervention expectations, and market-truth accuracy ($\text{Acc}^{\text{market}}$) as the accuracy on items whose ground truth aligns with market expectations. We further define $\text{Errors}^{\text{intervention}}$ and $\text{Errors}^{\text{market}}$ as the numbers of incorrect predictions whose predicted sign aligns with intervention and market expectations, respectively, and let $\text{Errors}^{\text{total}}$ denote the total number of errors in the ideology-contested subset. Their accuracy gap and directional error bias are then formalized as:

$$\Delta_{\text{acc}} = \text{Acc}^{\text{intervention}} - \text{Acc}^{\text{market}}, \quad B_{\text{dir}} = \frac{\text{Errors}^{\text{intervention}} - \text{Errors}^{\text{market}}}{\text{Errors}^{\text{total}}} \quad (2)$$

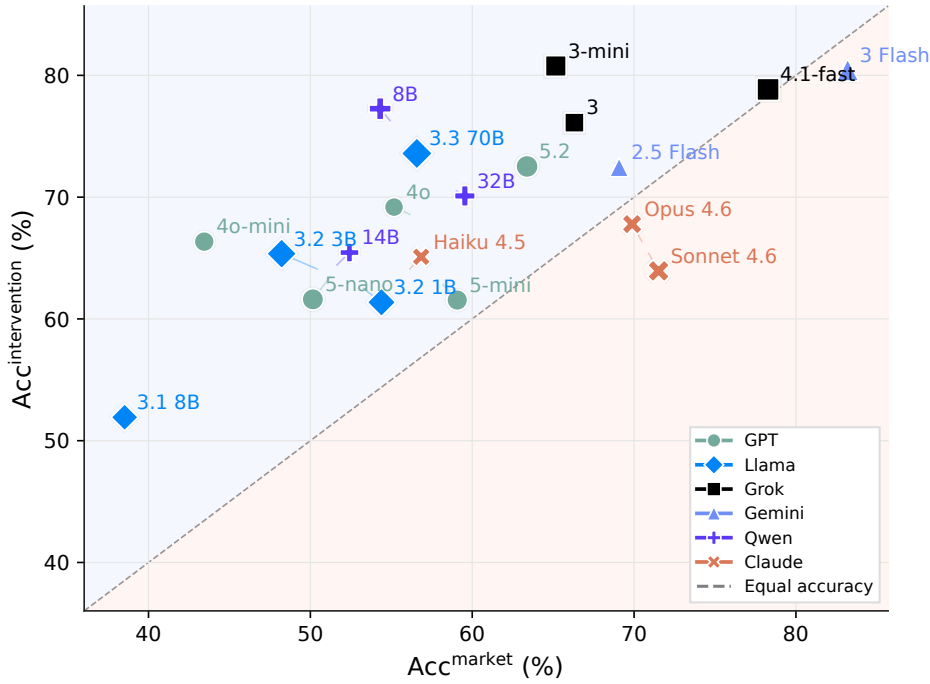


Figure 2: Accuracy on intervention-truth versus market-truth items across 20 LLMs. Most models fall above the equal-accuracy line, indicating a systematic advantage on intervention-aligned causal effects.

4.2 Main Results: Performance and Bias on Ideology-contested Items

Before examining accuracy, we note that the ground-truth labels themselves lean toward intervention-side (58.1% vs. 41.9% among directional items), and the models’ average predictions closely track this base rate (59.2% vs. 40.8%). Figure 2 reports model-level accuracy on the ideology-contested subset, focusing on intervention-truth versus market-truth items. The full table of main results is reported in Appendix B. We organize the findings along three dimensions: overall difficulty, asymmetric accuracy, and directional error bias.

Ideology-contested triplets are harder. Across all 20 models, accuracy on the ideology-contested subset is consistently lower than on the non-contested remainder. Closed-source models average 61.3% on ideology-contested items versus 74.5% on non-contested items, a gap of 13.2 percentage points (pp). Open-source models show a comparable pattern at a lower absolute level, averaging 52.5% versus 63.8% (11.3 pp gap). This uniform degradation indicates that economically contested causal relationships pose a systematically harder challenge for LLMs, independent of model family or scale.

Accuracy is asymmetric across ideological directions. Closed-source models average 72.9% on intervention-truth items but only 63.1% on market-truth items ($\Delta_{\text{acc}} = +9.7$ pp); open-source models show a wider gap at +15.1 pp. Of the 20 models, 18 exhibit a positive Δ_{acc} , with the largest gaps exceeding 20 pp (GPT-4o-mini, Qwen 3-8B, Llama 3.3-70B). The Claude family is a notable exception: Sonnet 4.6 and Opus 4.6 display a slight market-truth advantage ($\Delta_{\text{acc}} = -1.9$ and -0.6 , respectively).

Errors are systematically directional. The accuracy asymmetry is corroborated by the directional bias score B_{dir} . Closed-source models show a modest intervention-leaning bias on average ($B_{\text{dir}} = +2.9$), while open-source models exhibit a substantially stronger intervention skew ($B_{\text{dir}} = +8.8$). Among the 20 models, 15 exhibit $B_{\text{dir}} > 0$, with the largest directional biases observed in Llama 3.3-70B ($B_{\text{dir}} = +17.9$), Qwen 3-8B ($B_{\text{dir}} = +17.7$), and GPT-4o-mini ($B_{\text{dir}} = +15.5$). The consistent positive values of B_{dir} across the majority of models suggest that prediction errors on ideology-contested items are not uniformly distributed but instead reflect a systematic intervention-leaning tendency.

	Unmatched ($n_{\text{Intervention}} = 436$, $n_{\text{Market}} = 315$)	Matched ($n = 287$ per side)
$\text{Acc}_{\text{Intervention}} (\%)$	70.9	68.6
$\text{Acc}_{\text{Market}} (\%)$	59.3	58.0
$\Delta_{\text{acc}} (\text{pp})$	+11.5	+10.6
Models with $B_{\text{dir}} > 0$	18 / 20	18 / 20

Table 1: Accuracy gap between intervention-truth and market-truth items before and after difficulty matching. All accuracy values are 20-model means.

4.3 Heterogeneity Across Economic Subfields

To examine the fine-grained effects of ideological bias, we divide the directionally aligned subset into seven major economic subfields based on JEL (Journal of Economic Literature) classification codes, a standardized taxonomy maintained by the American Economic Association that assigns hierarchical alphanumeric labels to economic research topics (American Economic Association, 2024). Detailed mapping rules and subfield-level accuracy gaps are provided in Appendix A.

An intervention-truth advantage appears in six of seven subfields but is highly heterogeneous. The largest asymmetries arise in healthcare and welfare/redistribution, while labor and financial regulation exhibit more moderate gaps. Taxation is the sole exception, displaying a slightly reversed pattern. These results indicate that ideological distortion varies in both strength and direction by subfield rather than operating as a uniform leftward shift. The especially large intervention-truth advantage in healthcare ($\Delta_{\text{acc}} = +20.9$ pp) and welfare/redistribution ($\Delta_{\text{acc}} = +18.0$ pp) is consistent with LLMs responding more asymmetrically in policy domains where state intervention is ideologically salient. At the same time, exceptions such as taxation ($\Delta_{\text{acc}} = -1.0$ pp) caution against reducing the pattern to a blanket intervention bias. This heterogeneity clarifies that ideological distortion is concentrated in specific policy-relevant areas of economics rather than evenly distributed across all domains.

4.4 Robustness Check: Difficulty Matching

A natural concern is that market-truth items may be intrinsically harder, and that the accuracy gap in Figure 2 reflects this difficulty differential rather than ideological distortion. To rule out this confound, we scored each ideology-contested triplet on a 1–5 overall difficulty scale using GPT-5-mini and matched intervention-truth and market-truth triplets one-to-one within each *theme* $\times$ *difficulty* cell, yielding 287 triplets per side (details in Appendix C). As Table 1 shows, difficulty matching leaves every core finding virtually unchanged, with the same 18 of 20 models retaining $B_{\text{dir}} > 0$. The observed asymmetry is therefore not attributable to differential item difficulty. We further confirmed this result using a logistic regression on the matched sample with difficulty, subfield, and model fixed effects, where the intervention-truth indicator remained positive and highly significant.

5 Ideological Steering via In-Context Examples

The preceding analysis established that LLMs systematically favor intervention-truth causal triplets, with an average accuracy gap of $\Delta_{\text{acc}} = +11.5$ pp and intervention-leaning error directionality. In this section, we examine whether in-context examples—one of the simplest prompt-level interventions—can steer model predictions toward a target ideological direction.

5.1 Results

We compare four in-context conditions for each ideology-contested target triplet: no example (NONE; the baseline from Figure 2), a non-contested example (NON-CONTESTED), an intervention-aligned example (INTERVENTION-EX), and a market-aligned example (MARKET-EX). Each target is paired with examples sharing at least two JEL categories and a Jaccard context similarity ≥ 0.5 (full matching criteria in Appendix D.1). Our primary metric is the *Intervention–Market Example Gap*, defined as $\Delta_{\text{example}} = \text{Acc}_{\text{INTERVENTION-EX}} - \text{Acc}_{\text{MARKET-EX}}$. The results are shown in Table 2.

Intervention-truth targets show a consistent INTERVENTION-EX advantage. On intervention-truth targets, 17 of 20 models exhibit a positive Gap. The closed-source mean

Family	Model	Intervention-Truth Target					Market-Truth Target				
		None	Non-contested	Intervention-Ex	Market-Ex	Δ_{example}	None	Non-contested	Intervention-Ex	Market-Ex	Δ_{example}
Closed-Source Models											
OpenAI	GPT-4o-mini	69.5	47.5	50.3	58.2	-7.9	46.0	52.6	60.7	50.4	+10.3
	GPT-4o	72.5	43.4	45.5	43.7	+1.8	57.1	46.6	52.6	63.2	-10.6
	GPT-5-nano	66.7	55.7	55.9	53.8	+2.1	53.0	61.6	55.6	56.0	-0.4
	GPT-5-mini	65.6	68.7	74.1	63.9	+10.2	57.8	55.2	56.3	64.8	-8.5
	GPT-5.2	73.9	69.1	72.7	63.3	+9.4	63.2	59.8	68.1	66.4	+1.7
Claude	Haiku 4.5	67.4	70.4	72.0	70.9	+1.1	56.2	58.5	60.0	64.8	-4.8
	Sonnet 4.6	67.0	72.8	74.1	63.3	+10.8	68.9	72.3	71.1	63.2	+7.9
	Opus 4.6	67.7	82.0	77.6	70.9	+6.7	68.3	82.5	80.0	68.8	+11.2
Gemini	2.5 Flash	75.7	68.2	69.9	65.2	+4.7	65.4	62.8	61.5	67.2	-5.7
	3 Flash	82.6	87.5	83.9	80.4	+3.5	81.6	90.9	91.1	87.2	+3.9
Grok	3-mini	81.7	62.6	64.3	59.5	+4.8	64.1	67.8	71.1	69.6	+1.5
	3	77.3	63.2	60.8	66.5	-5.6	63.8	69.0	67.4	68.8	-1.4
	4-1 Fast	79.6	77.3	76.2	66.5	+9.8	74.9	65.6	74.1	73.6	+0.5
Closed-source mean		72.9	66.8	67.5	63.5	+4.0	63.1	65.0	66.9	66.5	+0.4
Open-Source Models											
Llama	3.1-8B	51.4	45.7	48.3	46.2	+2.0	41.9	47.7	51.9	44.8	-7.1
	3.2-1B	61.2	54.9	56.6	53.8	+2.8	50.8	60.8	52.6	49.6	+3.0
	3.2-3B	62.8	45.7	54.5	42.4	+12.1	49.2	50.5	57.0	44.8	+12.2
	3.3-70B	76.6	54.4	55.2	67.7	-12.5	56.2	61.2	62.2	68.0	-5.8
Qwen	3-8B	77.5	64.5	62.2	61.4	+0.8	55.9	61.6	65.2	64.0	-1.2
	3-14B	68.1	53.1	55.9	51.3	+4.7	52.4	47.3	48.1	66.4	-18.3
	3-32B	73.2	69.1	67.1	64.6	+2.6	58.7	50.1	49.6	61.6	-12.0
Open-source mean		67.3	55.3	57.1	55.3	+1.8	52.2	54.2	55.2	57.0	-1.8

Table 2: Accuracy (%) under different in-context example conditions.

Gap is $\Delta_{\text{example}} = +4.0$ pp, with the largest individual effects in GPT-5-mini ($\Delta_{\text{example}} = +10.2$ pp), Sonnet 4.6 ($\Delta_{\text{example}} = +10.8$ pp), and Grok 4-1 Fast ($\Delta_{\text{example}} = +9.8$ pp); open-source models show a smaller but still positive mean of $\Delta_{\text{example}} = +1.8$ pp.

The INTERVENTION-EX advantage persists even on market-truth targets. When the ground truth aligns with the market direction, 11 of 20 models still achieve higher accuracy under INTERVENTION-EX than under MARKET-EX. The closed-source mean Gap narrows to $\Delta_{\text{example}} = +0.4$ pp, while the open-source mean turns slightly negative ($\Delta_{\text{example}} = -1.8$ pp), indicating that the direction is model-family dependent.

5.2 Why Does the Gap Arise?

The Intervention-Market Example Gap is best understood as a target-dependent manifestation of the models’ underlying **intervention-oriented economic prior**. On intervention-truth targets, where this prior is aligned with the correct answer, predictions remain more stably anchored in an intervention-leaning direction, yielding a clear advantage for INTERVENTION-EX over MARKET-EX. This asymmetry is highly consistent across models: 17 of 20 show a positive Intervention-Market Example Gap on intervention-truth targets. On market-truth targets, by contrast, the prior conflicts with the ground truth, so the same asymmetry becomes weaker and less stable. Accordingly, only 9 of 20 models show higher accuracy under MARKET-EX than under INTERVENTION-EX, and the overall direction becomes model-family dependent.

This interpretation is further supported by the fact that the gap is not driven by clean positive steering from a single example. If in-context examples provided effective ideological steering, congruent examples—INTERVENTION-EX on intervention-truth targets and MARKET-EX on market-truth targets—should improve performance relative to the NONE baseline. Instead, the observed answer shifts are weak, uneven, and often fail to move in the corrective direction (Table 8 in Appendix D.2). Taken together, these results suggest that the asymmetric ICL pattern reflects the same intervention economic prior documented in the main results, while also showing that a single in-context example is insufficient to reliably override it.

5.3 Calibration Analysis

We examine how in-context examples (Intervention vs. Market) influence GPT-4o’s confidence distributions across ideological targets (Figure 3).

For intervention targets (left panel), MARKET-EX significantly increases confidence compared to INTERVENTION-EX ($\Delta\mu = +0.068$, $KS = 0.159$, $p = 0.039$). However, this elevated certainty reduces accuracy (43.7% vs. 45.5%, $\Delta_{\text{example}} = -1.8$ pp). This miscalibration

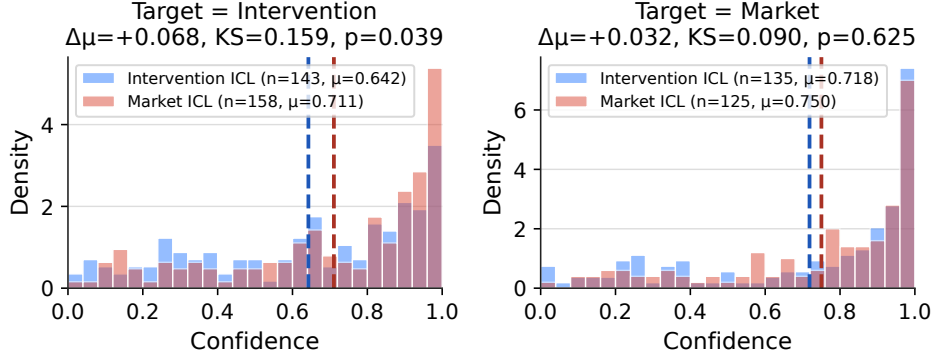
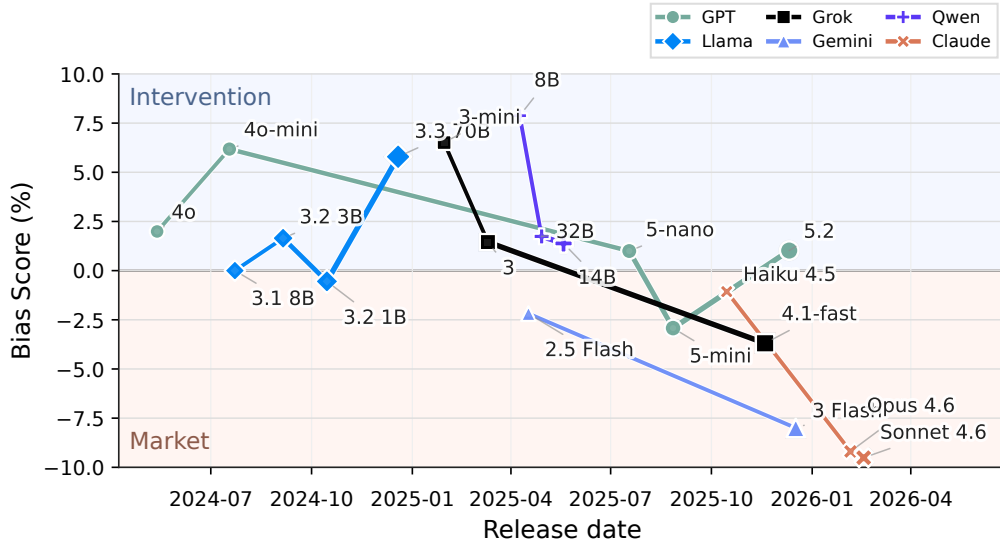


Figure 3: In-context learning confidence distribution by target ideology.

Figure 4: Directional bias score (B_{dir}) by model release date.

suggests that contrastive examples inflate certainty without correcting underlying judgments—problematically pairing high confidence with low accuracy.

Market targets (right panel) show a different pattern. The two confidence distributions largely overlap ($\Delta\mu = +0.032$, $KS = 0.090$, $p = 0.625$), yet the GPT-4o’s accuracy gap remains substantial ($\Delta_{\text{example}} = +10.6$ pp; 63.2% vs. 52.6%). Here, steering operates not by modulating confidence but by redirecting answer orientation—the model remains equally certain while arriving at different conclusions.

6 Discussion

6.1 Implications for LLM-Assisted Economic Analysis

Our results show that LLMs are not merely less accurate on ideologically contested economic causal questions; they are systematically more reliable in one ideological direction than the other. Across 18 of 20 models, accuracy is higher when the empirical sign aligns with intervention-oriented expectations than with market-oriented expectations, with average gaps of +9.7pp to +15.1pp. This makes the central concern not low accuracy alone, but directionally asymmetric reliability. Models can appear broadly competent while being substantially less dependable on one side of a contested economic question.

This asymmetry is qualitatively distinct from the opinion-level biases typically measured by political-compass-style instruments (Rozado, 2024; Hartmann et al., 2023; Santurkar et al.,

Model	Variant	Acc _{div}	Acc _{neu}	Δ_{acc}	B_{dir}
Qwen 3-14B	Base	44.3	53.7	+16.3	+6.0
	Instruct	52.2	69.3	+15.7	+8.0
	Δ	+7.9	+15.6	-0.5	+1.9
Qwen 3-8B	Base	46.3	55.7	+12.5	+3.2
	Instruct	57.5	71.6	+21.6	+17.7
	Δ	+11.2	+15.9	+9.1	+14.5

Table 3: Base vs. instruct comparison for Qwen pairs on the ideology-contested subset. In Δ_{acc} and B_{dir} , blue indicates intervention-leaning and red indicates market-leaning.

2023; Faulborn et al., 2025), which are designed to elicit expressed preferences rather than detect systematic error against verifiable ground truth. This distinction matters because reasoning bias is harder for users to detect: when a model predicts that a minimum wage increase sustains employment, users are likely to interpret the answer as an empirical summary rather than as the product of a skewed prior. Previous work also suggests that users may not spontaneously discount ideological content in LLM outputs (Potter et al., 2024). As a result, asymmetric reasoning errors may not only be difficult to detect in practice, but may also subtly influence downstream beliefs, judgments, and policy attitudes.

These findings imply that aggregate accuracy alone is an insufficient metric for evaluating LLMs in economic analysis. A model can achieve reasonable average performance while masking large directional disparities within ideologically contested cases. Our proposed metrics—the accuracy gap (Δ_{acc}) and directional bias score (B_{dir})—therefore provide a minimal but useful diagnostic for identifying failures that standard benchmarking would otherwise obscure. For LLM-assisted policy analysis, economic reporting, and decision support, such direction-aware evaluation should be treated as a basic robustness check rather than an optional extension.

6.2 Possible Sources of the Bias

It is not just a prompting artifact. Several results suggest that the observed bias is unlikely to be a shallow prompting artifact. The intervention-truth advantage remains after difficulty matching, indicating that the pattern is not simply driven by market-truth items being intrinsically harder. One-shot in-context steering also fails to reliably reverse the bias: although examples can shift answer tendencies in some cases, they do not consistently eliminate the intervention-truth advantage. Taken together, these findings point to a more persistent directional tendency in model behavior rather than a transient prompt-level effect.

Part of this directional tendency may originate earlier in the training pipeline. A potential source of this directional tendency may lie in pretraining, though subsequent alignment processes can further amplify it. The base-versus-instruct comparison supports this interpretation. As shown in Table 3, a noticeable intervention-truth advantage is already present at the base-model stage, before instruction tuning in both Qwen 3-14B and Qwen 3-8B. Instruction tuning improves overall accuracy, but it does not remove this asymmetry; in Qwen 3-8B, it substantially widens both the accuracy gap Δ_{acc} and the directional bias score B_{dir} . This pattern suggests that alignment does not simply wash out preexisting directional tendencies and may, in some cases, sharpen them, consistent with prior work showing that biases in pretraining corpora can propagate into downstream predictions (Feng et al., 2023).

Post-training pipelines may matter as well. The family-level patterns in Figures 2 and 4 suggest that the asymmetry cannot be reduced to a single uniform tendency. Claude Sonnet 4.6 and Opus 4.6 show a slight market-truth advantage, while within the GPT and Grok families the magnitude of the gap varies non-monotonically, only partially tracking scale-based explanations (Exler et al., 2025). These patterns suggest that family-specific post-training pipelines may amplify or suppress preexisting directional tendencies in different ways. Moreover, Figure 4 shows that this pattern is not static: earlier models cluster toward the intervention side, while the most recent releases trend toward market-truth accuracy, suggesting the bias is evolving alongside model generations. Since the alignment details of commercial models remain proprietary, however, disentangling these contributions remains an important direction for future work.

The distribution of ground-truth signs in the benchmark may also contribute to the bias. In the ideology-contested evaluation subset, the ground-truth sign in each triplet is anchored in the context-specific causal estimate supported by the paper’s rigorously designed empirical strategy. This is precisely why we focus on top-tier economics and finance journals: the goal is to evaluate model behavior against empirical claims that have survived unusually rigorous peer review and robustness scrutiny, not against arbitrary reported estimates. At the same time, rigor at the level of individual studies does not imply that the benchmarked distribution of signs perfectly matches the latent population distribution of true causal effects. Even when individual papers are carefully designed and peer-reviewed, the published literature may still reflect selection in which questions are studied, which results become visible and publishable, and which context-specific findings accumulate in the record. In ideologically contested domains, such selection and heterogeneity may themselves generate distributional skew in the benchmark.

Taken together, these results do not imply that the observed bias reflects ideological preference alone. A more cautious interpretation is that the pattern arises from an interaction among pretraining, alignment, family-specific post-training pipelines, and distributional regularities inherited from the empirical and discursive environments on which models are trained. Regardless of its precise source, however, the directional bias we document in LLM economic causal reasoning is itself policy-relevant.

6.3 Future Directions

A key next step is to distinguish more clearly between memory-based retrieval and mechanism-based reasoning. When models answer ideologically contested economic causal items correctly, they may do so either by recalling paper-specific empirical claims or by reasoning through competing economic mechanisms from the provided context. Disentangling these pathways would help clarify whether the bias we document primarily reflects memorized regularities in the literature, asymmetries in mechanism selection, or an interaction between the two. Future work should identify the sources of this asymmetry more precisely and develop better evaluation and mitigation strategies for directionally biased errors.

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A Economic Subfield Taxonomy

Ideology-contested causal triplets are grouped into seven economic subfields based on Journal of Economic Literature (JEL) classification codes. Table 4 presents the mapping rules and sample sizes.

Table 4: JEL code mapping to economic subfields.

Subfield	JEL Prefix	#Triplets	Definition of JEL code(s)
Taxation	H2*	73	Taxation, Subsidies, and Revenue
Healthcare	I1*	91	Health
Education	I2*	56	Education and Research Institutions
Welfare/Redistribution	H5*, I3*	87	National Government Expenditures and Related Policies; Welfare, Well-Being, and Poverty
Labor	J*	183	Labor and Demographic Economics
Financial Regulation	G*	172	Financial Economics
Trade	F1*	30	Trade
Other	—	175	Remaining codes (excluded)

A triplet may be associated with multiple JEL codes and therefore counted in multiple subfields. We use a vote-based counting scheme: each JEL code assigned to a triplet contributes one count to its corresponding subfield. For example, a triplet annotated with G35 and H25 contributes one count to *financial regulation* and one count to *taxation*. Codes outside the seven focal categories are labeled *other* and excluded from the subfield-level analysis.

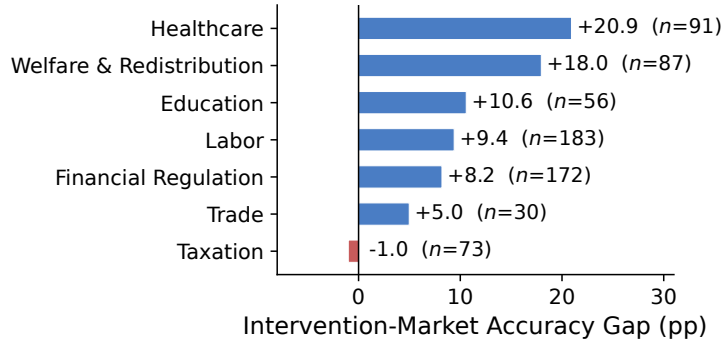


Figure 5: Mean accuracy gap of 20 models on Economic subfields

B Full Model-Level Results on Ideology-contested Items

This appendix reports the full model-level results underlying the aggregate patterns discussed in Section 4. While the main text focuses on the overall asymmetry across model groups, Table 5 provides the complete breakdown for each model on the ideology-contested subset.

For each model, we report accuracy on the non-contested subset, the ideology-contested subset, intervention-truth items, and market-truth items, along with two bias-oriented summary metrics. The *Accuracy Gap* is defined as $\Delta_{\text{acc}} = \text{Acc}_{\text{intervention}} - \text{Acc}_{\text{market}}$, so positive values indicate higher accuracy when the ground-truth sign aligns with intervention-oriented expectations, whereas negative values indicate the reverse. The *Error Direction Bias* measures whether incorrect predictions disproportionately align with intervention-oriented or market-oriented expectations; positive values indicate intervention-leaning errors, and negative values indicate market-leaning errors.

Consistent with the main findings, most models exhibit lower performance on ideology-contested items than on Non-contested items, and the majority show a positive intervention-

Family	Model	Accuracy (%)				Bias	
		Non-contested	Contested	Intervention	Market	Acc. Gap	Error Direction Bias
Closed-Source Models							
OpenAI	GPT-4o-mini	63.2	50.3	69.5	46.0	+23.5	+15.5
	GPT-4o	69.4	57.9	72.5	57.1	+15.3	+7.8
	GPT-5-nano	70.1	53.8	66.7	53.0	+13.7	+9.9
	GPT-5-mini	73.1	55.7	65.6	57.8	+7.8	+2.8
	GPT-5.2	75.8	63.4	73.9	63.2	+10.7	-2.6
Claude	Haiku 4.5	74.2	56.5	67.4	56.2	+11.2	+1.4
	Sonnet 4.6	76.3	63.0	67.0	68.9	-1.9	-8.7
	Opus 4.6	77.7	65.1	67.7	68.3	-0.6	-8.7
Gemini	2.5 Flash	75.9	61.8	75.7	65.4	+10.3	+6.5
	3 Flash	83.5	76.4	82.6	81.6	+1.0	-6.0
Grok	3-mini	76.7	64.1	81.7	64.1	+17.5	+13.5
	3	74.5	61.2	77.3	63.8	+13.5	+4.7
	4-1 Fast	78.2	67.5	79.6	74.9	+4.7	+1.8
Closed-source mean		74.5	61.3	72.9	63.1	+9.7	+2.9
Open-Source Models							
Llama	3.1-8B	49.3	42.7	51.4	41.9	+9.5	-1.0
	3.2-1B	56.4	48.3	61.2	50.8	+10.4	+1.9
	3.2-3B	56.7	48.9	62.8	49.2	+13.6	+6.8
	3.3-70B	72.1	58.0	76.6	56.2	+20.4	+17.9
Qwen	3-8B	71.6	58.5	77.5	55.9	+21.6	+17.7
	3-14B	69.3	53.5	68.1	52.4	+15.7	+8.0
	3-32B	71.1	57.8	73.2	58.7	+14.4	+10.1
Open-source mean		63.8	52.5	67.3	52.2	+15.1	+8.8

Table 5: Full model-level performance and ideological bias on the ideology-contested subset. Positive *Acc. Gap* indicates higher accuracy on intervention-truth items than on market-truth items, and positive *Error Direction Bias* indicates intervention-leaning errors.

market accuracy gap. This pattern is visible across both closed-source and open-source families, although its magnitude varies substantially by model family and scale. Notably, a small number of models—most prominently Claude Sonnet 4.6 and Claude Opus 4.6—show a slight market-truth advantage, highlighting that the asymmetry is strong but not universal.

C Difficulty Scoring, Matching, and Regression

Difficulty scoring. Each ideology-contested causal triplet was scored on a 1–5 difficulty scale by GPT-5-mini. The prompt instructed the model to evaluate five dimensions:

- **Domain knowledge:** depth of economics expertise required;
- **Context dependence:** reliance on the provided socio-economic context;
- **Ambiguity / confoundability:** plausibility of competing causal directions;
- **Causal reasoning complexity:** number of intermediate steps or mechanisms;
- **Evidence sufficiency:** whether the context provides enough information to determine the sign.

A composite *overall difficulty* score (1–5) was then derived from these five dimensions. The full scoring prompt is shown below.

Matching procedure. We matched intervention-truth and market-truth triplets one-to-one within each *theme* $\times$ *difficulty* cell. For JEL-based policy theme t and difficulty level d , we computed $m_{t,d} = \min(n_{t,d}^{\text{intervention}}, n_{t,d}^{\text{market}})$ and drew $m_{t,d}$ triplets per side. Cells where either side had zero count were excluded. The final matched sample contains 287 triplets per side (574 total); all 20 models’ prediction rows on these triplets were retained for re-evaluation.

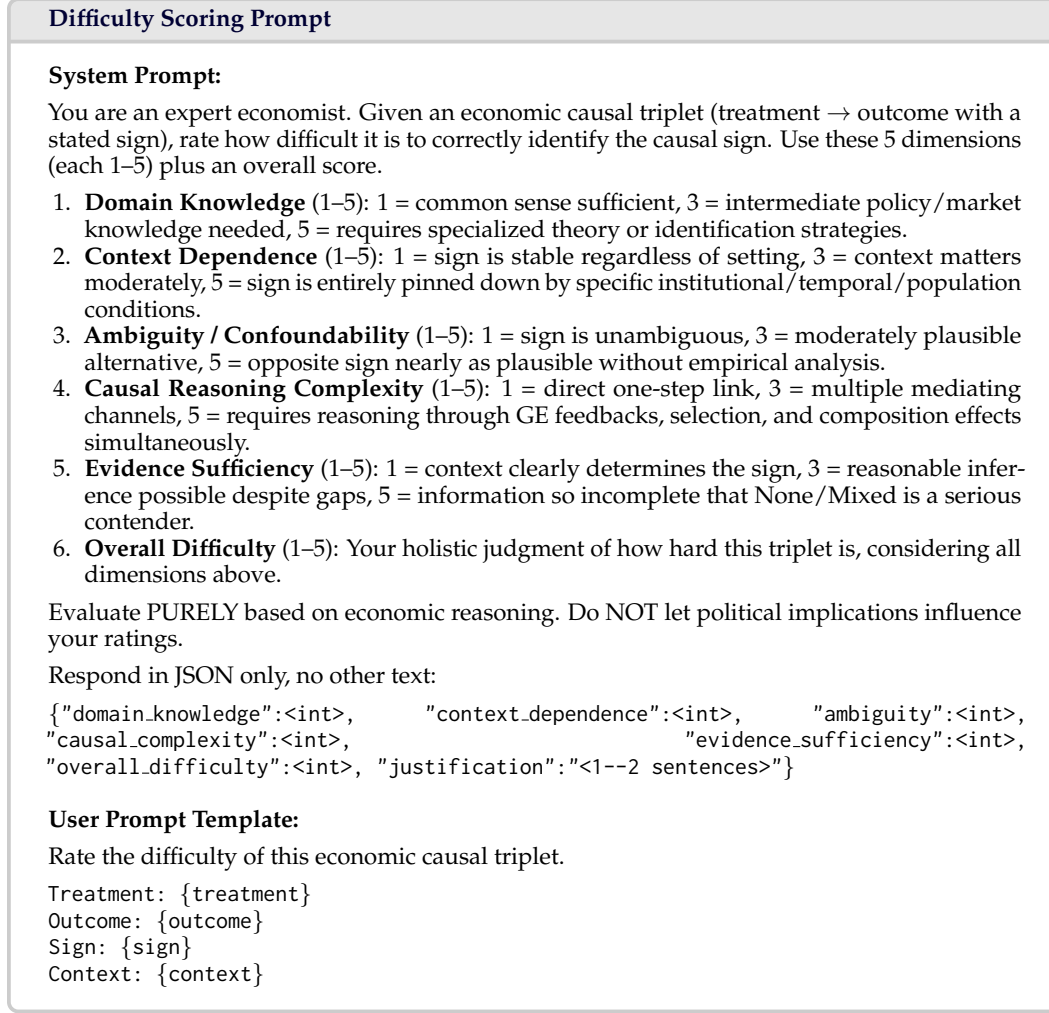


Figure 6: Prompt used for difficulty scoring (GPT-5-mini).

Logistic regression. As a complementary check, we estimated a logistic regression on the matched sample:

$$\text{logit}(P(\text{correct}_{i,m})) = \beta_0 + \beta_1 \mathbb{1}[\text{ground_truth} = \text{intervention}]_i + \gamma \mathbf{D}_i + \delta \mathbf{T}_i + \eta \mathbf{M}_m \quad (3)$$

where $\mathbf{D}_i$ denotes difficulty-level dummies, $\mathbf{T}_i$ theme fixed effects, and $\mathbf{M}_m$ model fixed effects. The coefficient of interest is β_1 , which captures the residual accuracy advantage of intervention-truth items after controlling for difficulty, subfield, and model heterogeneity.

Table 6 reports the result. The intervention-truth indicator is positive and highly significant ($\hat{\beta}_1 = 0.476$, $z = 4.66$, $p < 0.001$), corresponding to an odds-ratio increase of approximately 61% for correctly predicting intervention-truth items relative to market-truth items. No difficulty-level dummy reaches conventional significance, confirming that the matching procedure effectively balanced item difficulty.

	Coef.	Std. Err.	z	p
Ground-truth = intervention	+0.476	0.102	4.66	< 0.001
Difficulty = 3 (ref: 2)	+0.285	0.447	0.64	0.524
Difficulty = 4	+0.028	0.407	0.07	0.946
Difficulty = 5	−0.458	0.423	−1.08	0.279
Theme FE			Yes	
Model FE			Yes	

Table 6: Logistic regression on the difficulty-matched sample ($N = 11,480$). Only the variable of interest and difficulty controls are shown; model and theme fixed effects are included but omitted for brevity.

D ICL Example Details

D.1 Example Matching

For each ideology-contested target triplet, we define four prompting conditions: NONE, NON-CONTESTED, INTERVENTION-EX, and MARKET-EX. A candidate example is matched to a target if it satisfies the following conditions:

- the source papers of the target and example share at least two JEL codes; and
- their Jaccard similarity over JEL codes is at least 0.5.

Let W_t and W_e denote the sets of JEL codes assigned to the source papers from which the target and example triplets are extracted. We compute similarity as

$$J(W_t, W_e) = \frac{|W_t \cap W_e|}{|W_t \cup W_e|}, \quad |W_t \cap W_e| \geq 2, \quad J(W_t, W_e) \geq 0.5.$$

Table 7 reports the number of matched instances in each condition. The paired INTERVENTION-EX/MARKET-EX subset contains 125–158 instances per model.

Condition	Intervention-truth	Market-truth
NONE	436	315
NON-CONTESTED	1325	761
INTERVENTION-EX	143	135
MARKET-EX	158	125

Table 7: Number of matched instances per ICL condition and ground-truth alignment.

D.2 Steering Effects

Table 8 reports the mean answer shifts across 20 models when a single in-context example is introduced, relative to the NONE baseline. Congruent examples—INTERVENTION-EX on intervention-truth targets and MARKET-EX on market-truth targets—do not yield consistent corrective shifts, indicating that a single example is insufficient to reliably steer model predictions.

	Intervention-truth	Market-truth
INTERVENTION-EX: Δ intervention-aligned rate	−2.8	−0.8
MARKET-EX: Δ market-aligned rate	+3.1	−4.6

Table 8: Answer shifts from NONE (20-model mean, pp). Congruent conditions (top-left and bottom-right cells) show weak or negative shifts, suggesting that a single in-context example does not provide effective ideological steering.

E Prompts

Main Result Sign Prediction Prompt

Role:

You are an expert economist.

Task:

Given a context and a treatment–outcome pair, predict the most likely sign of the causal effect.

Sign Definitions:

- **+**: The treatment increases the outcome (positive and statistically significant effect).
- **-**: The treatment decreases the outcome (negative and statistically significant effect).
- **None**: No statistically significant effect.
- **mixed**: The effect varies across subgroups or specifications.

Prompt:

Context: {context}

Treatment--Outcome Pair

Treatment: {treatment}

Outcome: {outcome}

Output Format

Respond with a JSON object containing:

- predicted_sign: "+", "-", "None", or "mixed"
- reasoning: A concise explanation of your reasoning

Figure 7: Prompt used for main-result sign prediction.

Example-Augmented Sign Prediction Prompt

Role:

You are an expert economist.

Task:

You are given examples related to the target treatment–outcome pair, potentially with different causal signs. Predict the most likely sign for the treatment–outcome pair in the target context.

Sign Definitions:

- **+**: The treatment increases the outcome (positive and statistically significant effect).
- **-**: The treatment decreases the outcome (negative and statistically significant effect).
- **None**: No statistically significant effect.
- **mixed**: The effect varies across subgroups or specifications.

Prompt:

Reference Examples: {examples}

Target

Treatment: {treatment}

Outcome: {outcome}

Context: {context_new}

Output Format

Respond with a JSON object containing:

- predicted_sign: "+", "-", "None", or "mixed"
- reasoning: A concise explanation of your reasoning

Figure 8: Prompt used for example-augmented sign prediction.

Ideological Expected Sign Annotation Prompt**System Prompt:**

You are an annotation assistant for economics causal triplets. Your job is to read the given context, treatment, and outcome, and assign each ideology's expected causal sign.

Rules:

1. Output MUST be valid JSON only. No extra text.
2. Infer expected signs from ideological priors, NOT from the paper's empirical result.
3. Most causal triplets are NOT ideologically contested. If both ideologies would expect the same sign, assign the same sign to both.
4. Use null when no meaningful ideological prior exists for either side. Use the same non-null sign only when both sides would genuinely expect the same directional effect.
5. Also provide one brief reasoning string of at most 100 words explaining the sign judgment.

Theoretical Framework (based on Feldman & Johnston, 2014; Alesina & Giuliano, 2011)

Political ideology has at least two partially independent dimensions; here we focus ONLY on the **economic** dimension. The economic dimension captures attitudes toward the appropriate role of government versus the market in organizing economic life.

- **economic.conservative**: favors markets, limited government, lower taxes, deregulation, flexible labor markets, and skepticism toward redistribution; emphasizes incentives, efficiency, and moral hazard.
- **economic.liberal**: favors active government intervention, redistribution, regulation, labor protections, public services, and social insurance; emphasizes inequality, unequal opportunity, structural disadvantage, and market failure.

Sensitivity Criterion:

A triplet is ideologically contested ONLY when disagreement over the expected sign stems directly from the government-intervention-vs-market-freedom divide.

Do NOT assign different signs merely because:

- the topic is political or controversial,
- one ideology might normatively prefer the treatment or dislike the outcome,
- the setting involves firms, banks, or investors.

If no short, direct link from that divide to a different expected sign exists, assign null to both ideologies.

Task:

For each triplet, assign what each ideology would *expect* as the causal sign of treatment → outcome. Judge only the direction of the causal link.

Sign Options:

- **+**: positive causal effect
- **-**: negative causal effect
- **None**: no significant causal effect
- **Mixed**: mixed or context-dependent effects
- **null**: no ideological prior

Examples where signs differ:

- minimum wage increase → employment
- corporate tax rate → business investment

Examples where signs are the same:

- oil price shock → inflation
- R&D spending → firm productivity

Respond in JSON only, no other text:

```
{"ideological.expected.signs": {"economic.liberal.expected.sign": "<+|-|None|Mixed|null>",
"economic.conservative.expected.sign": "<+|-|None|Mixed|null>"}, "reasoning": "<brief reasoning, <=100 words>"}
```

User Prompt Template:

Paper Title: {title}
Context: {context}
Treatment: {treatment}
Outcome: {outcome}

Figure 9: Prompt used for ideological expected-sign annotation.